

1 Introduction

1.1 Motivation

2 Weak Orthogonality

2.1 Definitions

Definition 2.1 (Lifting problem). In any ambient category, a commuting square is a *lifting problem* when we ask whether there is a diagonal lift.

$$\begin{array}{ccc} A & \xrightarrow{h} & X \\ f \downarrow & \nearrow j & \downarrow g \\ B & \xrightarrow{k} & Y \end{array}$$

A *solution* to the lifting problem given by the square $(h, k) : f \rightarrow g$ is a morphism $j : B \rightarrow X$ making the triangles commute. That is, $j \circ f = h$ and $g \circ j = k$.

Definition 2.2 (Lifting property). Let $\mathcal{C}$ be a category and $\mathcal{M}, \mathcal{N}$ be classes of morphisms in $\mathcal{C}$. Let $f, g \in \mathcal{C}^{\rightarrow}$ be any morphisms.

- (i) We write $f \pitchfork g$ when every lifting problem of the following shape admits a solution.

$$\begin{array}{ccc} \bullet & \xrightarrow{\text{any}} & \bullet \\ f \downarrow & \nearrow \exists & \downarrow g \\ \bullet & \xrightarrow{\text{any}} & \bullet \end{array}$$

- (ii) We say that f *left lifts* against $\mathcal{N}$ and write $f \pitchfork \mathcal{N}$ when $f \pitchfork h$ for all $h \in \mathcal{N}$.

- (iii) We say that g *right lifts* against $\mathcal{M}$ and write $\mathcal{M} \pitchfork g$ when $k \pitchfork g$ for all $k \in \mathcal{M}$.

- (iv) We write $\mathcal{M} \pitchfork \mathcal{N}$ when every $k \in \mathcal{M}$ has left lifting against $\mathcal{N}$ and every $h \in \mathcal{N}$ has right lifting against $\mathcal{M}$. That is, every lifting problem of the following shape admits a solution.

$$\begin{array}{ccc} \bullet & \xrightarrow{\text{any}} & \bullet \\ \mathcal{M} \downarrow & \nearrow \exists & \downarrow \mathcal{N} \\ \bullet & \xrightarrow{\text{any}} & \bullet \end{array}$$

- (v) Let ${}^{\pitchfork}\mathcal{N}$ denote the class of morphisms which left lift against $\mathcal{N}$. Similarly, let $\mathcal{M}^{\pitchfork}$ denote the class of morphisms which right lift against $\mathcal{M}$.

Remark 2.3. That is, ${}^{\pitchfork}\mathcal{N}$ is the largest class of morphisms with left lifting against $\mathcal{N}$, so if $\mathcal{M} \pitchfork \mathcal{N}$, then $\mathcal{M} \subseteq {}^{\pitchfork}\mathcal{N}$. Symmetrically, $\mathcal{M}^{\pitchfork}$ is the largest class of morphisms with right lifting against $\mathcal{M}$, so if $\mathcal{M} \pitchfork \mathcal{N}$, then $\mathcal{N} \subseteq \mathcal{M}^{\pitchfork}$.

Definition 2.4 (Lifting structure). Let $\mathcal{C}$ be a category and $I : \mathbb{I} \rightarrow \mathcal{C}^\rightarrow$. Let $f : X \rightarrow Y$ be a morphism in $\mathcal{C}$. Then we have a functor $\psi : (\text{cod} \circ I)/X \rightarrow I/f$ sending the diagonal map below to the commutative square.

$$\begin{array}{ccc} A & \xrightarrow{\quad \quad \quad} & X \\ I(i) \downarrow & \nearrow & \downarrow f \\ B & \xrightarrow{\quad \quad \quad} & Y \end{array}$$

We say that $f : X \rightarrow Y$ has a $I^\square$ -structure if ψ has a section $\varphi : I/f \rightarrow (\text{cod} \circ I)/X$.

Concretely, this means that for any commutative square $(h, k) : I(i) \rightarrow f$ there is a chosen diagonal filler $\varphi(i, h, k)$, such that for any morphism $w : j \rightarrow i$ in I with $I(w) = (u, v)$, we have $\varphi(j, hu, kv) = \varphi(i, h, k) \circ v$.

$$\begin{array}{ccccc} A' & \xrightarrow{u} & A & \xrightarrow{h} & X \\ I(j) \downarrow & \nearrow \varphi(j, hu, vk) & \downarrow I(i) & \nearrow \varphi(i, h, k) & \downarrow f \\ B' & \xrightarrow{v} & B & \xrightarrow{k} & Y \end{array}$$

We write $f \in I^\square$ when f has a $I^\square$ -structure, where $I^\square$ is the full subcategory of $\mathcal{C}^\rightarrow$ of morphisms with $I^\square$ -structures.

Definition 2.5 (Weak factorization system). Let $\mathcal{L}, \mathcal{R}$ be classes of morphisms in $\mathcal{C}$. $(\mathcal{L}, \mathcal{R})$ is said to be a *weak factorization system* (wfs) if the following hold:

- (i) Every morphism in $\mathcal{C}$ factors into a map in $\mathcal{L}$ followed by a map in $\mathcal{R}$.
- (ii) $\mathcal{L} \pitchfork \mathcal{R}$, i.e., the classes are weakly orthogonal to one another.
- (iii) $\mathcal{L}^\pitchfork = \mathcal{R}$ and $^\pitchfork \mathcal{R} = \mathcal{L}$, and the classes $\mathcal{L}$ and $\mathcal{R}$ are closed under retracts.

2.2 Factorization

Proposition 2.6. Let $\mathcal{C}$ be a category, $I : \mathbb{I} \rightarrow \mathcal{C}^\rightarrow$ and $J : \mathbb{J} \rightarrow \mathcal{C}^\rightarrow$. If I factors through J , then $J^\square \subseteq I^\square$.

$$\begin{array}{ccc} \mathbb{I} & \xrightarrow{\quad L \quad} & \mathbb{J} \\ I \searrow & & \swarrow J \\ & \mathcal{C}^\rightarrow & \end{array}$$

Proof. Suppose $f : X \rightarrow Y$ has a $J^\square$ -structure $\varphi_J : J/f \rightarrow (\text{cod} \circ J)/X$. Then we can define an $I^\square$ -structure φ_I on f by taking $\varphi_I(i, h, k) = \varphi_J(Li, h, k)$, since $I(i) = J(Li)$. $\square$

Corollary 2.7. Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor. Suppose $I : \mathbb{I} \rightarrow \mathcal{C}^\rightarrow$ and $J : \mathbb{J} \rightarrow \mathcal{D}^\rightarrow$. Suppose $FI : \mathbb{I} \rightarrow \mathcal{D}^\rightarrow$ factors through J . Then $J^\square \subseteq (FI)^\square$.

$$\begin{array}{ccc} \mathbb{I} & \xrightarrow{\quad\quad\quad} & \mathbb{J} \\ I \downarrow & & \downarrow J \\ \mathcal{C}^\rightarrow & \xrightarrow{\quad F \quad} & \mathcal{D}^\rightarrow \end{array}$$

Proposition 2.8. Let $F \dashv G$ be an adjunction where $F : \mathcal{C} \rightarrow \mathcal{D}$. Let $\mathcal{S}$ be a collection of maps in $\mathcal{C}$. Let f be a morphism in $\mathcal{D}$. Then $F(\mathcal{S}) \pitchfork f$ iff $\mathcal{S} \pitchfork Gf$.

Proof. Let g be any morphism in $\mathcal{C}$. We have that $Fg \pitchfork f$ iff $g \pitchfork Gf$ by adjunction. $\square$

Proposition 2.9. Let $F \dashv G$ with $F : \mathcal{C} \rightarrow \mathcal{D}$. Suppose $I : \mathbb{I} \rightarrow \mathcal{C}^\rightarrow$. Then $(FI)^\square = G^{-1}(I^\square)$.

Proof. Let f be a morphism in $\mathcal{D}$ such that Gf has a $I^\square$ -structure φ . Then define a $(FI)^\square$ -structure ρ on f by taking $\rho(i, h, k) = \varphi(i, \tilde{h}, \tilde{k})$, where $(\tilde{h}, \tilde{k}) : I(i) \rightarrow f$ is the transpose of $(h, k) : FI(i) \rightarrow f$ given by the adjunction. Similarly, given that f has an $(FI)^\square$ -structure, we can define an $I^\square$ -structure on Gf . $\square$

Proposition 2.10. Let $I : \mathbb{I} \rightarrow \mathcal{C}$ and $(\mathcal{L}, \mathcal{R})$ be a wfs on $\mathcal{C}$ such that $\mathcal{R} = I^\square$. Let $J : \mathbb{J} \rightarrow \mathcal{D}$ and $(\mathcal{L}', \mathcal{R}')$ be a wfs on $\mathcal{D}$ such that $\mathcal{R}' = J^\square$. Suppose $F : \mathcal{C} \rightarrow \mathcal{D}$ is such that FI factors through J . Then $F(\mathcal{L}) \subseteq \mathcal{L}'$.

2.10 yields a fact that can be used to show a functor is left Quillen.

Corollary 2.11. Let $\mathcal{C}$ and $\mathcal{D}$ be model categories. Let $I : \mathbb{I} \rightarrow \mathcal{C}^\rightarrow$ be such that $I^\square = \text{Fib}_{\mathcal{C}}$ and let $J : \mathbb{J} \rightarrow \mathcal{D}^\rightarrow$ be such that $J^\square = \text{Fib}_{\mathcal{D}}$. Suppose $F : \mathcal{C} \rightarrow \mathcal{D}$ is such that FI factors through J . Then $F(\text{TCof}_{\mathcal{C}}) \subseteq \text{TCof}_{\mathcal{D}}$.

3 Motivating Example

3.1 Cube categories

Definition 3.1 (Cartesian cube category, $\mathbb{C}$). The *Cartesian cube category*, denoted $\mathbb{C}$, is the Lawvere theory with constants 0, 1, and no other operations (the theory of bipointed objects, the initial Lawvere theory with two constants.) A substitution

$$(x_1, \dots, x_m \mid a_1, \dots, a_n)$$

is a morphism in $\mathbb{C}$ if a_i is a term in the language with only constants 0 and 1, which means precisely that $a_i \in \{0, x_1, \dots, x_m, 1\}$.

Definition 3.2 (Dedekind cube category, $\mathbb{D}$). The *Dedekind cube category*, denoted $\mathbb{D}$, is the Lawvere theory of bounded distributive lattices. It has two binary operations, $\wedge$ (meet) and $\vee$ (join), called *connections*, which are both commutative, associative, idempotent and subject to these additional axioms:

- identity: $a \vee 0 = a$ and $a \wedge 1 = a$
- absorption: $a \vee (a \wedge b) = a$ and $a \wedge (a \vee b) = a$
- distributivity: $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$ and $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$

A substitution $(x_1, \dots, x_m \mid a_1, \dots, a_n)$ is a morphism in $\mathbb{D}$ if a_i is a term in this larger language. Two morphisms $(x_1, \dots, x_m \mid a_1, \dots, a_n)$, $(x_1, \dots, x_m \mid b_1, \dots, b_n)$ are identified whenever a_i and b_i are provably equal under the axioms.

Remark. In $\mathbb{C}$ and $\mathbb{D}$, we have $[n] = [1]^n$ as they are Lawvere theories.

Construction 3.3 (Canonical inclusions). We have an identity-on-objects inclusion

$$j : \mathbb{C} \rightarrow \mathbb{D}$$

sending a morphism $(x_1, \dots, x_m \mid a_1, \dots, a_n) : [m] \rightarrow [n]$ in $\mathbb{C}$ to the morphism presented by the same substitution (now seen as terms in the expanded language with connections) in $\mathbb{D}$. This assignment is faithful because different variables and constants cannot become provably equal, even under the presence of connections.

Remark. $\mathbb{D}$ can be any of the other cartesian cubical categories. Any Lawvere theory with constants 0 and 1 will work here, including join semi-lattices or cubes with negations. We use the Dedekind cube category for $\mathbb{D}$ with the aim of constructing a model structure on $\widehat{\mathbb{D}}$ which presents spaces.

Construction 3.4. j^* and its left and right Kan extensions along Yoneda embedding form an adjoint triple.

Proposition 3.5. j^* is conservative. In particular, it is faithful.

Proof. The canonical inclusion $j : \mathbb{C} \rightarrow \mathbb{D}$ is identity on objects, in particular surjective on objects. Hence, the restriction j^* is conservative. $\square$

3.2 Geometric morphisms

Proposition 3.6. The adjunction $j^* \dashv j_*$ is a geometric morphism $\widehat{\mathbb{C}} \rightarrow \widehat{\mathbb{D}}$.

Proposition 3.7. Any flat distributive lattice is strictly bipointed.

Proposition 3.8. The adjunction $j_! \dashv j^*$ is a geometric morphism $\widehat{\mathbb{D}} \rightarrow \widehat{\mathbb{C}}$.

3.3 Left Quillen

Proposition 3.9. $j^* : \widehat{\mathbb{D}}_{\text{int}} \rightarrow \widehat{\mathbb{C}}_{\text{int}}$ is left Quillen.

Proof. $\widehat{\mathbb{D}}_{\text{int}} = \widehat{\mathbb{D}}_{\text{cis}}$ since δ is a retract of an anodyne cofibration in $\widehat{\mathbb{D}}$. Anodyne cofibrations in $\widehat{\mathbb{C}}$ are trivial cofibrations in $\widehat{\mathbb{C}}_{\text{int}}$. Any $s : A \rightarrow B$ exhibiting A as a strong deformation retract of B is anodyne. $j^* \delta_0 : 1 \rightarrow j^* J$ is a strong deformation retract of $j^* J$, hence it is a trivial cofibration in $\widehat{\mathbb{C}}_{\text{int}}$. Then $j^* : \widehat{\mathbb{D}}_{\text{cis}} \rightarrow \widehat{\mathbb{C}}_{\text{int}}$ is left Quillen. Since $\widehat{\mathbb{D}}_{\text{int}} = \widehat{\mathbb{D}}_{\text{cis}}$, j^* is left Quillen between the interval model structures. $\square$

Proposition 3.10. $\text{id} : \widehat{\mathbb{C}}_{\text{int}} \rightarrow \widehat{\mathbb{C}}_{\text{eqv}}$ is left Quillen.

Proof. $\widehat{\mathbb{C}}_{\text{eqv}}$ has more trivial cofibrations. (find section in ACCRS) [ACC⁺25] $\square$

Proposition 3.11. $\widehat{\mathbb{D}}_{\text{int}}$ and $\widehat{\mathbb{D}}_{\text{eqv}}$ coincide.

Proof. Cavallo-Sattler (find section) [CS25] $\square$

Proposition 3.12. $j^* : \widehat{\mathbb{D}}_{\text{eqv}} \rightarrow \widehat{\mathbb{C}}_{\text{eqv}}$ is left Quillen.

Proof. By 3.9 and 3.10, $j^* = \text{id} \circ j^*$ is left Quillen from $\widehat{\mathbb{D}}_{\text{int}}$ to $\widehat{\mathbb{C}}_{\text{eqv}}$, as it is a composite of left Quillen functors

$$\widehat{\mathbb{D}}_{\text{int}} \xrightarrow{j^*} \widehat{\mathbb{C}}_{\text{int}} \xrightarrow{\text{id}} \widehat{\mathbb{C}}_{\text{eqv}}$$

Then use the fact that $\widehat{\mathbb{D}}_{\text{int}} = \widehat{\mathbb{D}}_{\text{eqv}}$, as in 3.11. $\square$

3.4 Right Quillen

Proposition 3.13. The following commutes:

$$\begin{array}{ccc}
 \widehat{\mathbb{C}}^\Sigma & \xrightarrow{j!} & \widehat{\mathbb{D}}^\Sigma \\
 \uparrow \Delta & & \uparrow \Delta \\
 \widehat{\mathbb{C}} & \xrightarrow{j!} & \widehat{\mathbb{D}}
 \end{array}$$

Proposition 3.14. $j!(\mathrm{TCof}_{\widehat{\mathbb{C}}^\Sigma}) \subseteq \mathrm{TCof}_{\widehat{\mathbb{D}}^\Sigma}$, and consequently, $j!(\mathrm{TCof}_{\widehat{\mathbb{C}}}) \subseteq \mathrm{TCof}_{\widehat{\mathbb{D}}}$. (because f equivariant fibration iff Δf fibration in cubical species)

Proposition 3.15. $j^* : \widehat{\mathbb{D}}_{\mathrm{eqv}} \rightarrow \widehat{\mathbb{C}}_{\mathrm{eqv}}$ is right Quillen.

4 Local Model Structure

Homotopical preliminaries

For this section, suppose that we work in an arbitrary model category.

Lemma 4.1. Let $f : X \rightarrow Y$ be any fibration. Then there is a fibration $f' : X' \rightarrow Y'$ where X' and Y' are fibrant replacements of X and Y respectively.

$$\begin{array}{ccc} X & \xrightarrow{\quad} & X' \\ \downarrow & & \downarrow \\ Y & \xrightarrow{\quad} & Y' \end{array}$$

Lemma 4.2. Suppose we have a commutative cube such that the front, back, and right sides are homotopy pullback squares. Then so is the left side.

Lemma 4.3. Homotopy does not depend on choice of a cylinder object. Let X and Y be any objects, and suppose Z is a cylinder object for X .

$$X + X \rightrightarrows Z \rightarrow X$$

Let $f, g : X \rightrightarrows Y$ be parallel arrows and $h : Z \rightarrow Y$ be a homotopy between f and g .

$$\begin{array}{ccc} X & & \\ \downarrow & \searrow f & \\ Z & \xrightarrow{h} & Y \\ \uparrow & \nearrow g & \\ X & & \end{array}$$

Suppose Z' is another cylinder object for X . Then we can find a homotopy $h' : Z' \rightarrow Y$ between f and g .

Lemma 4.4. A weak equivalence between fibrations is a homotopy equivalence.

Lemma 4.5. **Quillen equivalence fact** If $F \dashv G$ is Quillen with G preserving and reflecting weqs and the unit is pointwise weq then Quillen equiv.

4.1 Setup

For the rest of the section, we work in the following situation.

Let $F \dashv G \dashv H$ be a triple of adjoints between model categories $\mathcal{C}$ and $\mathcal{D}$ with $G : \mathcal{D} \rightarrow \mathcal{C}$. We shall also assume:

- (a) G is both left and right Quillen.
- (b) Cofibrations in $\mathcal{C}$ and $\mathcal{D}$ are monomorphisms. In particular, all objects are cofibrant in both $\mathcal{C}$ and $\mathcal{D}$.
- (c) For all (cofibrant) object $X \in \mathcal{C}$, $X \rightarrow GFX$ is a weak equivalence. For all fibrant $X \in \mathcal{C}$, $GHX \rightarrow X$ is a weak equivalence.
- (d) G is conservative, in particular faithful, and the unit $1 \rightarrow HG$ is pointwise monic.

Notation. Let W denote the class $G^{-1}[\text{Weq}(\mathcal{C})]$, and F_W denote the class of maps with right lifting property against $\text{Cof}(\mathcal{D}) \cap W$.

Proposition 4.6. $(\text{Cof}(\mathcal{D}), W, F_W)$ is a model structure.

We will shortly prove Prop 4.6 in §4.2.

Notation. Let $\mathcal{D}_W$ be the model category described by $(\text{Cof}(\mathcal{D}), W, F_W)$. To distinguish the model categories $\mathcal{D}$ and $\mathcal{D}_W$, we will sometimes say *local* fibrations or *local* fibrant objects to mean fibrations and fibrant objects in $\mathcal{D}_W$.

4.2 Existence of model structure

Observation. Since G is left Quillen, it preserves weak equivalences between cofibrant objects. Since all objects in $\mathcal{D}$ are cofibrant, $\text{Weq}(\mathcal{D}) \subseteq W$. Also, because G is a right adjoint and faithful, G preserves and reflects monomorphisms. Thus, $G^{-1}[\text{TCof}(\mathcal{C})] = \text{Cof}(\mathcal{D}) \cap W$.

Observation. W satisfies 3-for-2 as $\mathcal{C}$ is a model category. Since G is right Quillen, W is stable under homotopy pullbacks.

Observation. Since $\mathcal{D}$ is a model category, $(\text{Cof}(\mathcal{D}), \text{TFib}(\mathcal{D}))$ is already a weak factorization system. Hence it suffices to check that $(\text{Cof}(\mathcal{D}) \cap W, F_W)$ is a weak factorization system and that $F_W \cap W = \text{TFib}(\mathcal{D})$.

Lemma 4.7. $H[\text{Fib}(\mathcal{C})] \subset F_W$.

Proof. Let f be a fibration in $\mathcal{C}$. We want Hf to have right lifting against $\text{Cof}(\mathcal{D}) \cap W$. Suppose $c : A \rightarrow B$ is in $\text{Cof}(\mathcal{D}) \cap W$, which means Gc is a trivial cofibration in $\mathcal{C}$. Then $Gc \pitchfork f$ since f is a fibration in $\mathcal{C}$, which is equivalent to $c \pitchfork Hf$ by adjunction. $\square$

Lemma 4.8. Let $f : X \rightarrow Y$ be any fibration. Let $P \rightarrow Y$ be the pullback of HGf along the unit map $Y \rightarrow HGY$.

$$\begin{array}{ccc}
 X & \xrightarrow{\quad} & HGX \\
 \downarrow f & \searrow & \uparrow \\
 & P & \\
 \downarrow & \swarrow & \downarrow HGf \\
 Y & \xrightarrow{\quad} & HGY
 \end{array}$$

Then the comparison map $X \rightarrow P$ is in W .

Proof. The simple case is when $f : X \rightarrow Y$ is a fibration between fibrant objects. Since X and Y are fibrant, we have that the unit maps $X \rightarrow HGX$ and $Y \rightarrow HGY$ are in W . Consider the unit square $f \rightarrow HGf$ and the pullback $P \rightarrow Y$.

$$\begin{array}{ccc}
 X & \xrightarrow{w} & HGX \\
 \downarrow & \searrow & \uparrow w \\
 & P & \\
 \downarrow & \swarrow & \downarrow \\
 Y & \xrightarrow{w} & HGY
 \end{array}$$

Then $P \rightarrow HGX$ is in W since it is a pullback of $Y \rightarrow HGY$ which is in W . We deduce that $X \rightarrow P$ is in W by 3-for-2 of W .

Now, suppose that f is any fibration. Let $f' : X' \rightarrow Y'$ be a replacement of f by a fibration between fibrant objects.

$$\begin{array}{ccc}
 X & \xrightarrow{\quad} & X' \\
 \downarrow & & \downarrow \\
 Y & \xrightarrow{\quad} & Y'
 \end{array}$$

Consider the cube formed by the unit squares for f and f' . Let $P \rightarrow Y$ and $P' \rightarrow Y'$ be the

pullbacks of HGf and HGf' along the corresponding unit maps.

$$\begin{array}{ccccc}
 X & \xrightarrow{\quad} & HGX & \xrightarrow{\quad} & HGX' \\
 \downarrow \scriptstyle f & \searrow \scriptstyle \circ & \downarrow \scriptstyle \circ & \searrow \scriptstyle \circ & \downarrow \scriptstyle \circ \\
 & P & & P' & \\
 \downarrow \scriptstyle f & \swarrow \scriptstyle \circ & \downarrow \scriptstyle \circ & \swarrow \scriptstyle \circ & \\
 Y & \xrightarrow{\quad} & HGY & \xrightarrow{\quad} & HGY' \\
 \downarrow \scriptstyle f & \searrow \scriptstyle \circ & \downarrow \scriptstyle \circ & \searrow \scriptstyle \circ & \\
 & Y' & & &
 \end{array}
 \tag{1}$$

In the diagram, arrows in W are marked with a circle and weak equivalences are marked with a bar. Left face of the cube (red) is a homotopy pullback. Since H and G are right Quillen, right face of the cube (blue) is also a homotopy pullback [??] since the vertical maps are fibrations. Then so is the square formed between $P \rightarrow Y$ and $P' \rightarrow Y'$. By pasting lemma, we deduce that the square formed by $X \rightarrow P$ and $X' \rightarrow P'$ is also a homotopy pullback. Since $X' \rightarrow P'$ is in W , so is $X \rightarrow P$. $\square$

Lemma 4.9. $(\text{Cof}(\mathcal{D}) \cap W, F_W)$ is a weak factorization system.

Proof. We first show that any fibration $f : X \rightarrow Y$ can be factored into a map in $\text{Cof}(\mathcal{D}) \cap W$ followed by a map in F_W .

$$\begin{array}{ccc}
 X & \xrightarrow{\quad} & HGX \\
 \downarrow \scriptstyle f & \searrow \scriptstyle w & \downarrow \scriptstyle HGf \\
 & P & \\
 \downarrow \scriptstyle f & \swarrow & \downarrow \scriptstyle HGf \\
 Y & \xrightarrow{\quad} & HGY
 \end{array}$$

By the previous lemma, $X \rightarrow P$ is in W . $X \rightarrow P$ is a monomorphism since the unit map $X \rightarrow HGX$ is a monomorphism and factors through it. Hence $X \rightarrow P \rightarrow Y$ is a factorization of f into a map in $\text{Cof}(\mathcal{D}) \cap W$ followed by a map in F_W .

If $f : A \rightarrow X$ is any map, then we can first factor it into a trivial cofibration followed by a fibration. Since $\text{TCof}(\mathcal{D}) \subseteq \text{Cof}(\mathcal{D}) \cap W$, we finish by factoring the fibration part as above.

It remains to check that $\text{Cof}(\mathcal{D}) \cap W = {}^h(F_W)$. Since $\text{Cof}(\mathcal{D}) \cap W = G^{-1}[\text{TCof}(\mathcal{C})]$, it is enough to show that if a cofibration $c : A \rightarrow B$ has left lifting against F_W , then Gc is a trivial

cofibration in $\mathcal{C}$. We make use the fact that $H[\text{Fib}(\mathcal{C})] \subset F_W$: $c \pitchfork F_W$ implies $c \pitchfork H[\text{Fib}(\mathcal{C})]$. By adjunction, $c \pitchfork H[\text{Fib}(\mathcal{C})]$ iff $Gc \pitchfork \text{Fib}(\mathcal{C})$. That is, Gc is a trivial cofibration. $\square$

Lemma 4.10. $F_W \cap W = \text{TFib}(\mathcal{D})$.

Proof. Let $f : X \rightarrow Y$ be in $F_W \cap W$. Then we can factor f into a cofibration followed by a trivial fibration using the model structure on $\mathcal{D}$. Since $\text{Weq}(\mathcal{D}) \subseteq W$, the cofibration is in W by 3-for-2. Then f is a retract of the trivial fibration by the retract lemma, since it right lifts against $\text{Cof}(\mathcal{D}) \cap W$. $\square$

Observation. $\text{id} : \mathcal{D} \rightarrow \mathcal{D}_W$ is left Quillen.

Proposition 4.11. $F \dashv G$ is a Quillen equivalence between $\mathcal{C}$ and $\mathcal{D}_W$.

Proof. G preserves and reflects weak equivalences and unit maps $X \rightarrow GFX$ are weak equivalences $[\text{??}]$. $\square$

4.3 Local fibrations

Suppose these additional hypotheses:

- (e) The underlying 1-categories of $\mathcal{C}$ and $\mathcal{D}$ are Grothendieck 1-toposes.
- (f) $\mathcal{C}$ and $\mathcal{D}$ both have universes for fibrations, Frobenius property (and therefore right proper), equivalence extension property, and fibration extension property.

The goal of this section is to prove:

Proposition 4.12. $\mathcal{D}_W$ has universes for fibrations, Frobenius property, equivalence extension property, and fibration extension property.

Lemma 4.13. Let $f : A \rightarrow X$ be any map. f is a homotopy equivalence in $\mathcal{D}_W$ if and only if f is a homotopy equivalence in $\mathcal{D}$.

Proof. A cylinder object in $\mathcal{D}$ is still a cylinder object in $\mathcal{D}_W$. $\square$

Lemma 4.14. Let $f_0 : X_0 \rightarrow Y$ and $f_1 : X_1 \rightarrow Y$ be weakly equivalent fibrations over Y . Then f_0 is local if and only if f_1 is.

Lemma 4.15. Let Y be a fibrant object in $\mathcal{D}$; $f : X \rightarrow Y$ and $f' : X' \rightarrow Y$ be in F_W ; and $g : X \rightarrow X'$ is such that $f = f'g$. Then g is in W iff it is a weak equivalence.

$$\begin{array}{ccc} X & \xrightarrow{g} & X' \\ & \searrow f & \swarrow f' \\ & Y & \end{array}$$

Proof. Since f and f' are fibrations in $\mathcal{D}_W$ and g a weak equivalence in $\mathcal{D}_W$, g must be a homotopy equivalence in $\mathcal{D}_W$ and therefore in $\mathcal{D}$. $\square$

Proposition 4.16. Let f be any fibration in $\mathcal{D}$. f is in F_W if and only if the unit square $f \rightarrow HGf$ is a homotopy pullback in $\mathcal{D}$.

Proof. ($\Rightarrow$) Suppose f is in F_W . Let $P \rightarrow Y$ be the pullback of HGf along $Y \rightarrow HGY$. The comparison map $X \rightarrow P$ is in W by Lemma 4.8. Since both f and $P \rightarrow Y$ are in F_W , $X \rightarrow P$ is a weak equivalence.

($\Leftarrow$) Suppose $f \rightarrow HGf$ is a homotopy pullback square. Since X and P are both homotopy pullbacks, the comparison map $X \rightarrow P$ is a weak equivalence in $\mathcal{D}$. Since $X \rightarrow P$ is also monic, it is a trivial cofibration. Then f is a retract of $P \rightarrow Y$ which is in F_W . $\square$

Notation. Define a functorial factorization on $\mathcal{D}$ as follows. Given any $f : X \rightarrow Y$, f can be factored into $X \xrightarrow{\lambda f} Ef \xrightarrow{\rho f} Y$

$$\begin{array}{ccc}
 X & \xrightarrow{\eta_X} & HGX \\
 \downarrow f & \searrow \lambda f & \uparrow \\
 & Ef & \\
 \uparrow \rho f & \nearrow & \downarrow HGf \\
 Y & \xrightarrow{\eta_Y} & HGY
 \end{array}$$

where $\rho f : Ef \rightarrow Y$ is the pullback of HGf along η_Y and $\lambda f : X \rightarrow Ef$ is the comparison map.

Lemma 4.17. This functorial factorization is cartesian.

Proof. Suppose we have a pullback square $f \rightarrow f'$. By pasting lemma, it is enough to show that $\rho f \rightarrow \rho f'$ is a pullback. Since G and H are right adjoints, $HGf \rightarrow HGf'$ is also a pullback square. ρf and $\rho f'$ are pullbacks of HGf and HGf' along η_Y and $\eta_{Y'}$.

$$\begin{array}{ccccc}
 Ef & \xrightarrow{\quad} & HGX & & \\
 \downarrow \rho f & \searrow & \downarrow & \searrow & \\
 & Ef' & \xrightarrow{\quad} & HGX' & \\
 & \downarrow \rho f' & \downarrow HGf & \downarrow HGf' & \\
 Y & \xrightarrow{\quad} & HGY & \xrightarrow{\quad} & HGY' \\
 & \searrow & \downarrow & \searrow & \\
 & Y' & \xrightarrow{\quad} & HGY' &
 \end{array}$$

Since the back, front, and right faces of the cube are pullbacks, so is the left face. $\square$

Proposition 4.18. Let $f : X \rightarrow Y$ be any fibration in $\mathcal{D}$. f is in F_W if and only if $\lambda f : f \rightarrow \rho f$ has a retraction.

Proof. λf is in $\text{Cof}(\mathcal{D}) \cap W$ and since f is a fibration, ρf is in F_W . If f is in F_W , then it right lifts against λf . Then by the retract lemma λf has a retraction. On the other hand, if λf has a retraction, then f is in F_W since F_W is closed under retracts. $\square$

Observation. Let f be in F_W . Then λf is a weak equivalence by 4.16.

Lemma 4.19. Let $f' : X' \rightarrow Y'$ be in F_W . Suppose $f : X \rightarrow Y$ is a pullback of $f' : X' \rightarrow Y'$ along a monomorphism $Y \rightarrow Y'$.

$$\begin{array}{ccc} X & \xrightarrow{\quad} & X' \\ f \downarrow & \lrcorner & \downarrow f' \\ Y & \xrightarrow{\quad} & Y' \end{array}$$

Let $r : Ef \rightarrow X$ be a retraction of λf over Y . Then there is a retraction $r' : Ef' \rightarrow X'$ of $\lambda f'$ over Y' such that r is the pullback of r' along $X \rightarrow X'$.

$$\begin{array}{ccc} X & \xrightarrow{i} & X' \\ r \nearrow \downarrow \lambda f & & \lambda f' \downarrow \searrow r' \\ Ef & \xrightarrow{\quad} & Ef' \\ \rho f \downarrow & & \downarrow \rho f' \\ Y & \xrightarrow{\quad} & Y' \end{array}$$

Proof. Since f' and f are both in F_W , $\lambda f'$ and λf are both trivial cofibrations. Let Q be the pushout of λf along i . Since $\lambda f \rightarrow \lambda f'$ is a pullback, $Q \rightarrow Ef'$ is a monomorphism as Q is a union of subobjects.

$$\begin{array}{ccc} X & \xrightarrow{i} & X' \\ \lambda f \downarrow & \nearrow & \downarrow \lambda f' \\ & Q & \\ Ef & \xrightarrow{\quad} & Ef' \end{array}$$

The map $X' \rightarrow Q$ is a trivial cofibration since it is a pushout of λf . Then $Q \rightarrow Ef'$ is a

trivial cofibration by 3-for-2. Since f' is a fibration, it right-lifts against $Q \rightarrow Ef'$.

$$\begin{array}{ccc}
 Q & \xrightarrow{\langle \text{id}_{X'}, ir \rangle} & X' \\
 \downarrow & \nearrow \text{dashed} & \downarrow f' \\
 Ef' & \xrightarrow{\rho f'} & Y'
 \end{array}$$

Take $r' : Ef' \rightarrow X'$ to be a solution to the lifting problem. □

Lemma 4.20. For each $f : X \rightarrow Y$ in $\mathcal{D}$, there is a morphism $\text{Retr}_\lambda(f) \rightarrow Y$ whose sections correspond to retractions of $\lambda f : f \rightarrow \rho f$.

Proof. Using the lcc structure, we can define $[\rho f, f] : D_1 \rightarrow Y$ and $[f, f] : D_2 \rightarrow Y$.

$$\begin{array}{ccccc}
 \text{Retr}_\lambda(f) & \longrightarrow & D_1 & & \\
 \downarrow & & \downarrow - \circ \lambda f & \searrow [\rho f, f] & \\
 Y & \xrightarrow{\text{id}_X} & D_2 & \searrow [f, f] & \\
 & \searrow 1_Y & & & Y
 \end{array}$$

Let $\text{Retr}_\lambda(f)$ be the pullback of the cospan. Elements of the fiber of $- \circ \lambda f : [\rho f, f] \rightarrow [f, f]$ over $\text{id}_X : f \rightarrow f$, and thus sections of $\text{Retr}_\lambda(f) \rightarrow Y$, correspond to retractions of λf . □

Lemma 4.21. Let $f : X \rightarrow Y$ and $g : Y' \rightarrow Y$ be in $\mathcal{D}$. Then $\text{Retr}_\lambda(g^*f) \rightarrow Y'$ is the pullback of $\text{Retr}_\lambda(f) \rightarrow Y$ along g , i.e., $\text{Retr}_\lambda(g^*f) = g^*(\text{Retr}_\lambda(f))$.

$$\begin{array}{ccc}
 \text{Retr}_\lambda(g^*f) & \longrightarrow & \text{Retr}_\lambda(f) \\
 \downarrow & \lrcorner & \downarrow \\
 Y' & \xrightarrow{g} & Y
 \end{array}$$

Proof. The functorial factorization $f \mapsto (\lambda f, \rho f)$ is cartesian, so $\lambda(g^*f) = g^*(\lambda f)$ and $\rho(g^*f) = g^*(\rho f)$. The result follows as $g^* : \mathcal{D}/Y \rightarrow \mathcal{D}/Y'$ is a cartesian closed functor. □

Lemma 4.22. Let $f : X \rightarrow Y$ be any map. Let $f' : X' \rightarrow Y' = \text{Retr}_\lambda(f)$ be the pullback of f along $g : \text{Retr}_\lambda(f) \rightarrow Y$. Then $\text{Retr}_\lambda(f') \rightarrow Y'$ has a canonical section.

Proof. That is, $f' = g^* f$ so by 4.21, $\text{Retr}_\lambda(f') = g^*(\text{Retr}_\lambda(f))$.

$$\begin{array}{ccccc}
 X' & \xrightarrow{\quad} & X & & \\
 \downarrow f' & & \downarrow f & & \\
 \text{Retr}_\lambda(f') & \xrightarrow{\quad} & \text{Retr}_\lambda(f) & \xrightarrow{g} & Y \\
 & \searrow & \downarrow g & & \\
 & & \text{Retr}_\lambda(f) & \xrightarrow{g} & Y
 \end{array}$$

The canonical section $\text{Retr}_\lambda(f) \rightarrow g^*(\text{Retr}_\lambda(f))$ is induced by the identity on $\text{Retr}_\lambda(f)$. $\square$

Proposition 4.23. $\mathcal{D}_W$ has universes for fibrations. That is, there are universes for local fibrations.

Proof. Let $\pi : \dot{U} \rightarrow U$ be a universe for fibrations in $\mathcal{D}$. Define $U_W := \text{Retr}_\lambda(\pi)$, and let $\pi_W : \dot{U}_W \rightarrow U_W$ be the pullback of π along $U_W \rightarrow U$. By 4.22, π_W is a local fibration.

$$\begin{array}{ccc}
 \dot{U}_W & \longrightarrow & \dot{U} \\
 \pi_W \downarrow & \lrcorner & \downarrow \pi \\
 U_W & \longrightarrow & U
 \end{array}$$

We claim that $\pi_W : \dot{U}_W \rightarrow U_W$ is the universe for local fibrations. We have to check

- (i) Given a local fibration $f : X \rightarrow Y$, we can find $\ulcorner f \urcorner : Y \rightarrow U_W$ such that f is the pullback of π_W along $\ulcorner f \urcorner$.
- (ii) U_W satisfies realignment.

Let $f : X \rightarrow Y$ be a local fibration. Then in particular f is a fibration, so it is classified by some $\ulcorner f \urcorner : Y \rightarrow U$ such that $f = \ulcorner f \urcorner^*(\pi)$.

$$\begin{array}{ccccc}
 & & \dot{U}_W & \longrightarrow & \dot{U} \\
 & & \downarrow \pi_W & & \downarrow \pi \\
 X & \xrightarrow{\quad} & U_W & \longrightarrow & U \\
 \downarrow f & & \downarrow \ulcorner f \urcorner & & \\
 Y & \xrightarrow{\quad} & & &
 \end{array}$$

For (i), it suffices to show that $\ulcorner f \urcorner$ factors through U_W . Since f is a local fibration, there is a section $Y \rightarrow \text{Retr}_\lambda(f)$ and thus a $Y \rightarrow U_W$ making the bottom triangle commute. As $f = \ulcorner f \urcorner^*(\pi)$, $\text{Retr}_\lambda(f)$ is the pullback of U_W along $\ulcorner f \urcorner$.

Let $f : X \rightarrow Y$ be a local fibration and $f' : X' \rightarrow Y'$ be the pullback of f along some monomorphism $m : Y' \rightarrow Y$. Suppose $f' = g^* \pi_W$ for some $g : Y' \rightarrow U_W$. We want to find $h : Y \rightarrow U_W$ such that $g = hm$ and $f = h^* \pi_W$.

$$\begin{array}{ccccc}
 X' & \xrightarrow{\quad} & U_W & & \\
 \downarrow f' & \searrow & \downarrow \pi_W & \nearrow & \\
 & X & & & \\
 & \downarrow f & & & \\
 Y' & \xrightarrow{g} & U_W & & \\
 \downarrow m & \searrow & \downarrow h & \nearrow & \\
 & Y & & &
 \end{array}$$

Let g' be the composite $Y' \xrightarrow{g} U_W \rightarrow U$. By pasting lemma, f' is the pullback of π along this map. Since $\pi : \dot{U} \rightarrow U$ has realignment property, we can find $k : Y \rightarrow U$ such that $g' = km$ and $f = k^* \pi$.

$$\begin{array}{ccccccc}
 X' & \xrightarrow{\quad} & U_W & \xrightarrow{\quad} & \dot{U} & & \\
 \downarrow f' & \searrow & \downarrow \pi_W & \nearrow & \downarrow \pi & & \\
 & X & & & & & \\
 & \downarrow f & & & & & \\
 Y' & \xrightarrow{g} & U_W & \xrightarrow{\quad} & U & & \\
 \downarrow m & \searrow & \downarrow h & \nearrow & \downarrow k & & \\
 & Y & & & & &
 \end{array}$$

As f is a local fibration, k factors through U_W via some $h : Y \rightarrow U_W$. Then $f : X \rightarrow Y$ is a pullback of π_W along h . □

Proposition 4.24. $\mathcal{D}_W$ has the Frobenius property.

Proof. We have to show that the class $\text{TCof}(\mathcal{D}_W) = \text{Cof}(\mathcal{D}) \cap W$ is stable under pullbacks along local fibrations. Let $c : A \rightarrow B$ be in $\text{Cof}(\mathcal{D}) \cap W$ and $g : B' \rightarrow B$ be a local fibration.

$$\begin{array}{ccc}
 A' & \xrightarrow{\quad} & A \\
 \downarrow g^*c & \lrcorner & \downarrow c \\
 B' & \xrightarrow{g} & B
 \end{array}$$

We need to show that g^*c is in $\text{Cof}(\mathcal{D}) \cap W$. Since $\text{Cof}(\mathcal{D}) \cap W = G^{-1}[\text{TCof}(\mathcal{C})]$, we show that $G(g^*c)$ is a trivial cofibration in $\mathcal{C}$. G preserves pullbacks and (all) fibrations in $\mathcal{D}$, so Gg is

a fibration in $\mathcal{C}$ and $G(g^*c) = (Gg)^*(Gc)$. We have $Gc \in \text{TCof}(\mathcal{C})$ since $c \in \text{Cof}(\mathcal{D}) \cap W$.

$$\begin{array}{ccc} GA' & \longrightarrow & GA \\ G(g^*c) \downarrow & \lrcorner & \downarrow Gc \\ GB' & \xrightarrow{Gg} & GB \end{array}$$

$\mathcal{C}$ has the Frobenius property, giving us $G(g^*c) \in \text{TCof}(\mathcal{C})$. □

Proposition 4.25. $\mathcal{D}_W$ has the equivalence extension property.

Proof. If a map $w : f_0 \rightarrow f_1$ between local fibrations f_0 and f_1 over an object A is in W , then it is a homotopy equivalence in both $\mathcal{D}_W$ and $\mathcal{D}$. Therefore it is a weak equivalence in $\mathcal{D}$. Suppose we have a cofibration $c : A \rightarrow B$ and a local fibration g_1 over B such that $f_1 = c^*g_1$. By the equivalence extension property for $\mathcal{D}$, we have a weak equivalence $z : g_0 \rightarrow g_1$ extending w where $g_0 \in \text{Fib}(\mathcal{D})$. But g_0 is local if and only if g_1 is. □

Proposition 4.26. $\mathcal{D}_W$ has the fibration extension property.

Proof. Let $c : A \rightarrow B$ be in $\text{TCof}(\mathcal{D}_W) = \text{Cof}(\mathcal{D}) \cap W$ and $p : X \rightarrow A$ be a local fibration. We have to find a local fibration $q : Y \rightarrow B$ extending p , i.e. p is the pullback of q along c .

$$\begin{array}{ccc} X & \xrightarrow{\dots\dots\dots} & Y \\ p \downarrow & \lrcorner & \downarrow q \\ A & \xrightarrow[c]{W} & B \end{array}$$

Gc is a trivial cofibration and Gp is a fibration (in $\mathcal{C}$). Using the fibration extension property for $\mathcal{C}$, we can find $f : Z \rightarrow GB$ such that $Gp = (Gc)^*f$.

$$\begin{array}{ccc} GX & \xrightarrow{\quad\quad\quad} & Z \\ Gp \downarrow & \lrcorner & \downarrow f \\ GA & \xrightarrow[Gc]{\text{TCof}(\mathcal{C})} & GB \end{array}$$

H preserves pullbacks and fibrations, so $HGp = (HGc)^*(Hf)$.

$$\begin{array}{ccc} HGX & \longrightarrow & HZ \\ HGp \downarrow & \lrcorner & \downarrow Hf \\ HGA & \xrightarrow{HGc} & HGB \end{array}$$

Let $p_0 : P \rightarrow A$ be the pullback of HGp along $\eta_A : A \rightarrow HGA$ and q_0 be the pullback of Hf

along $\eta_B : B \rightarrow HGB$. We obtain the following commuting cube:

$$\begin{array}{ccccc}
 P & \xrightarrow{\quad} & Q & & \\
 \downarrow p_0 & \searrow & \downarrow q_0 & \searrow & \\
 & HGX & \xrightarrow{\quad} & HZ & \\
 & \downarrow HGp & & \downarrow Hf & \\
 A & \xrightarrow{\quad c \quad} & B & & \\
 \searrow \eta_A & & \searrow \eta_B & & \\
 & HGA & \xrightarrow{HGc} & HGB &
 \end{array}$$

Note that vertical maps of the cube are local fibrations. Since the front, left, and right faces of the cube are pullbacks, so is the back. That is, $p_0 = c^*q_0$.

$$\begin{array}{ccccc}
 X & \xrightarrow{\quad} & Y & & \\
 \searrow p & & \searrow q & & \\
 & P & \xrightarrow{\quad} & Q & \\
 & \downarrow p_0 & & \downarrow q_0 & \\
 & A & \xrightarrow{\quad c \quad} & B &
 \end{array}$$

Recall that $p : X \rightarrow A$ is a local fibration. Then the comparison map $X \rightarrow P$ is a weak equivalence. Then by 4.25 (equivalence extension property for $\mathcal{D}_W$) we can find a local fibration $q : Y \rightarrow B$ such that $p = c^*q$. $\square$

References

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